

1 Solve.

$$4c - 9 = 13$$

$$4c = 13 + 9$$

$$\frac{4}{4} c = \frac{22}{4}$$

$$c = 5.5$$

$$c = \dots\dots\dots 5.5 \dots\dots\dots [2]$$

2 Work out.

$$\frac{16.71 + 46.13}{\sqrt{8.6^2 - 3.5^2}}$$

Give your answer correct to 2 significant figures.

$$= 7.9994$$

$$= 8.0$$

$$\dots\dots\dots 8.0 \dots\dots\dots [2]$$

3 In the USA, one gallon of fuel costs \$4.83 .
In the UK, one litre of fuel costs £1.62 .

The exchange rate is £1 = \$1.215 .

1 gallon = 3.785 litres

In which country does 1 litre of fuel cost more and by how much?
Give your answer in dollars.

UK : £1.62 fuelUSA

$$£1 = 1.215$$

$$\frac{1 \text{ gallon } \$4.83}{3.785} = \frac{3.785 \text{ L} = \$4.83}{3.785 \quad 8.985}$$

$$£1.62 = x$$

$$1 \text{ L} = \$1.78$$

$$x = 1.9683$$

$$x = 1.97 \$$$

$$\dots\dots\text{UK}\dots\dots \text{by } \$ \dots\dots\dots 0.69 \dots\dots\dots [3]$$